

Survival Analysis for Wildfire Evacuation: Predicting Time-to-Hit Using Early-Stage Data

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Wildfire emergency response requires not only identifying whether a fire may threaten an evacuation zone, but also estimating the risk of fire hit within some specific time window. Using the WiDS wildfire dataset, we modeled conditional time-to-hit risk from a 5-hour landmark based on the first five hour fire characteristic. Our goals were to identify early-stage wildfire characteristics associated with evacuation-zone exposure and to estimate hit probabilities at 12, 24, 48, and 72 hours after ignition. After feature screening and Cox diagnostics, proportional hazards violations motivated the use of an Accelerated Failure Time model as the primary inferential framework, with Random Survival Forest used for sensitivity analysis. The final model showed that centroid movement speed, fire-start day of week, and the observation process were more informative for short-horizon time-to-hit than static baseline characteristics such as initial fire area. Mean 3-fold cross-validated concordance of the AFT model was 0.755, and full-sample apparent AUC exceeded 0.80 across all horizons. These findings suggest that early-stage wildfire dynamics can support useful urgent fire identification, thereby promoting the efficiency of resource allocation.

Wildfire Evacuation | Accelerated Failure Time Model | Random Survival Forest

As extreme weather events become more frequent, wildfires can cause not only ecological damage but also billions of dollars in losses and human casualties (1). In practice, emergency response requires timely and reliable forecasts to support evacuation planning and resource allocation. For decision-making, it is not enough to estimate whether a fire may threaten an evacuation zone or not; responders also need to know when that threat is likely to reach.

Existing wildfire studies have made important progress in modeling occurrence and spatial propagation. They have used trigger point and fire-spread models to support evacuation planning (2, 3). However, these physical models are often computationally expensive and demand detailed spatial inputs and environmental data. More importantly, they are less focused on probabilistic time-to-hit prediction. Wildfire data also exhibit substantial right censoring, since some fires are controlled before reaching evacuation zones and others may never reach within the observed window due to weather, terrain and vegetation impact. Survival analysis provides a natural framework for this situation, since it models time-to-event outcomes with censoring and enables dynamic risk estimation for time-sensitive actions.

Our research objective is to identify early-stage wildfire characteristics associated with the hazard of reaching an evacuation zone and to estimate the probability of a fire hitting an evacuation zone within 12, 24, 48, and 72 hours of initial detection, while evaluating model performance and calibration across all four prediction horizons.

Based on prior studies, we expected that fires with larger initial size, faster early growth, faster centroid movement, and movement more aligned toward evacuation zones would reach evacuation zones within a shorter time. We also expected temporal context and observation quality to influence estimated risk.

Using a landmark survival framework, we built an accelerated failure time model and found that movement-related dynamics, fire-start time, and the observation process were more informative for short-term time-to-hit than static geometric characteristics.

Literature Review

Statistical modeling has played an increasingly important role in wildfire risk assessment, shifting from deterministic models to probabilistic models that support decision-making. Early statistical approaches focused on modeling fire ignition as a spatial-temporal point process, often using logistic regression or generalized additive models to estimate occurrence probabilities based on environmental and human-related predictors. In more recent work, wildfire study has extended to stochastic models and different outcomes such as fire duration, size, and growth (4).

Survival analysis has recently been applied to wildfire processes that involve duration outcomes. A study in Ontario demonstrated the use of standard survival analysis methods, including Kaplan-Meier estimators, accelerated failure time models, and Cox proportional hazards models, to study forest fire control time (5). Their work showed how local weather, fuel moisture, fire size at the onset of fire attack, and fire management are associated with the time until a fire is brought under control, illustrating the interpretability of survival models for time-to-event outcomes in the context of wildfire. Further work introduced a new two-stage approach to estimate the effects of fire management and environmental factors on fire-size growth (6). They substituted the conventional time dimension with fire size, treating the size at initial assessment as a left-truncation point and the size at “being held” (with no further increase in size) as the terminal event. Their work demonstrated that weather and fuel type significantly influence fire expansion, suggesting the possibility of modeling spatial growth using time-to-event frameworks.

Besides analyzing the suppression and control of wildfire using survival framework, prior work has also examined evacuation-related wildfire risk through trigger-point and warning-time modeling, highlighting the importance of operation when fire conditions may threaten communities or evacuation zones (2, 3). However, these approaches do not directly estimate conditional time-to-hit using early-stage wildfire features using survival analysis methods.

Research Gap. Prior work pays limited attention to short-term time-to-hit prediction for evacuation zones and makes limited use of

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Y.Z. conducted introduction, exploratory data analysis, sensitivity analysis, and discussion. Y.W. led methodology design and developed the primary AFT modeling framework, conducted the primary model. Both authors contributed to interpretation of results, and revision of the manuscript.

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early-stage wildfire features for emergency ranking. This project addresses that gap by modeling conditional 12-, 24-, 48-, and 72-hour time-to-hit risk using early-stage wildfire features based on a landmark design.

Materials and Methods

This section describes the data structure, feature engineering pipeline, the primary AFT modeling strategy, and the sensitivity analyses used to assess robustness.

Data Description. In our study, we used the WiDS wildfire dataset, with each wildfire event as the unit of analysis. This dataset contained 221 wildfire events, including 69 hit events and 152 right-censored observations (68.8% censoring rate) (7).

The study used a landmark design with a 0–5 hour feature window and a 5–72 hour follow-up window as shown in Figure 1. Predictors were constructed from early-stage fire behavior during the first 5 hours after initial detection (34 characteristics in total), including fire growth, centroid kinematics, directional features relative to nearby evacuation zones, temporal metadata, and observation-quality variables. Follow-up time begins at the 5-hour landmark, and the outcome is defined within a time-to-event framework. An event is recorded ($= 1$) if the fire comes within 5 km of an evacuation zone during follow-up; otherwise, the observation is treated as right-censored ($= 0$). The variable `time_to_hit_hours` measures the additional time from the landmark until the fire reaches within 5 km of an evacuation zone. For censored observations, the variable `time_to_hit_hours` is the last available follow-up time within the 72-hour window. Thus, all predicted survival probabilities are conditional on the fire having survived event-free through the initial 5-hour feature window. Accordingly, evaluation horizons of 12, 24, 48, and 72 hours after ignition correspond to 7, 19, 43, and 67 hours from the landmark, respectively.

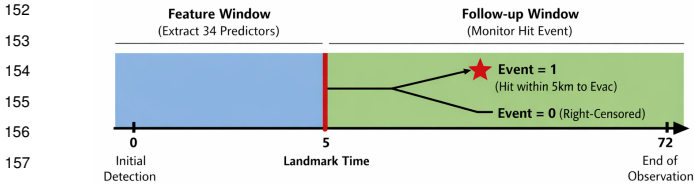


Fig. 1. Landmark time design showing the 0–5 hour feature window, the 5-hour landmark, and follow-up for evacuation-zone hit or right-censoring.

Exploratory Data Analysis and Feature Engineering. Exploratory data analyses were used to motivate preprocessing choices, guide feature reduction, and support construction of the primary regression model pipeline.

Exploratory analyses motivating pre-processing. Exploratory data analysis and plots were used to guide preprocessing. Histograms of fire-start hour and day-of-week suggested cyclical pattern rather than monotonic structure. We therefore encoded these variables using sine and cosine pairs, which preserve circular continuity at the boundary (e.g., 23:00 and 00:00 remain close in this form) and avoid arbitrary cutoffs. Several wildfire-dynamics variables showed strong right-skewness in violin and box plots, suggesting log transformation. In the stratified Kaplan–Meier curves, low resolution observations have consistently higher survival probabilities, supporting inclusion of observation quality as a control

variable. Seasonal stratification also suggested higher hazard during summer months, motivating a binary summer indicator. Finally, numeric predictors were standardized using z-scores.

Definition-based feature screening. We first removed distance-based and distance-derived predictors, because the event outcome itself was defined by distance to evacuation zones, reducing the feature pool from 34 to 23. We then removed variables replaced by engineered cyclical encodings (including temporal and directional variables), duplicate representations of the same quantity at different scales (e.g., raw and log-transformed versions), and additional observation-process variables already summarized by the low-resolution flag, retaining a 17-variable exploratory set.

For the primary interpretable model, we further reduced this set to a 12-variable candidate subset. This reduction was guided by interpretability, reduction of redundant feature representations, and closer alignment with the research question of wildfire movement toward evacuation zones. First, we removed absolute spread-bearing variables in favor of directional alignment relative to evacuation zones, since absolute geographic direction is difficult to interpret when the broader geographic context of the sampled fires is unavailable. Second, we retained speed-based rather than displacement-based kinematic measures, while speed more directly reflects dynamic fire movement. Third, within highly related feature families, we preferred relative representations than absolute versions that were easier to interpret.

Penalized Cox regression. The final feature selection stage uses an Elastic Net Cox model to mitigate overfitting and multicollinearity. We set $\alpha = 0.5$ and choose λ by 5-fold cross-validation with the C-index criterion. This step reduces the set from 12 to 9 covariates. The resulting 9-variable set was then put into an initial multivariable Cox proportional hazards model (Cox PH) for fitting and diagnostic assessment. Since proportional hazards assumption diagnostics indicated violations for several covariates, Cox PH was not retained as our primary inferential model; instead, we adopted an Accelerated Failure Time (AFT) framework.

Collinearity diagnostics and model refinement. While fitting the initial 9-variable AFT model (see description of this model in Primary Model Section), severe collinearity was detected between radial growth and centroid movement speed. We compared reduced models retaining only one of the two dynamic-related predictors using AIC, BIC, and 3-fold CV C-index. Since removing either covariate had minimal impact on discrimination and fit, we retained centroid speed as the more interpretable kinematic predictor, yielding a **final 8-variable AFT model**.

The final 8-variable set includes one observation quality variable (a binary low resolution flag), one fire geometry variable (initial area), one fire kinematic variable (centroid movement speed), one directional alignment variable (alignment angle cosine), and four fire-start time (cosine-encoded hour, sine/cosine-encoded day-of-week, and summer indicator). Brief definitions of the final predictors are summarized in Table 9 in the Appendix.

Primary Model. As indicated by the Schoenfeld Residuals test (score test), four of all covariates violate the proportional hazards assumption. We therefore considered several alternatives within the survival modeling framework to relax the PH assumption. The first was stratification or adding time-varying coefficients. However, stratification cannot estimate the effect of the stratified variables themselves, and we had four variables violating the

assumption, stratifying all of them would result in a substantial loss of interpretability. Alternatively, adding time-varying coefficient terms for all violating variables would make the model too complex. We also considered a discrete-time survival model, but since our time-to-hit outcome is continuous, artificially discretizing the time axis would introduce arbitrary choices and potential loss of precision. For these reasons, we chose the AFT model, which does not rely on the proportional hazards assumption and can directly model log survival time (8). Beyond these statistical considerations, the AFT formulation is suitable for wildfire modeling: its multiplicative acceleration structure reflects how early-stage fire characteristics such as growth rate, spread speed, and directionality can impact the speed at which a fire approaches the evacuation zone. Therefore, we fitted an AFT model including all the main effects of the 8 covariates listed in the previous section. This model works by regressing log survival time on a linear predictor η_i . Specifically, we have

$$\begin{aligned}\eta_i = & \beta_0 + \beta_1 \log_area_first_i + \beta_2 \log_centroid_speed_i \\ & + \beta_3 \text{alignment_cos}_i + \beta_4 \text{hour_cos}_i \\ & + \beta_5 \text{low_temporal_resolution}_i + \beta_6 \text{dow_sin}_i \\ & + \beta_7 \text{dow_cos}_i + \beta_8 \text{is_summer}_i\end{aligned}$$

so that the AFT model: $\log(T_i) = \eta_i + \epsilon_i$ where ϵ is independent of all covariates and has mean zero.

Under the AFT formulation, the survival function can be written as

$$S(t | X_i) = S_0(t \exp(-\eta_i))$$

where $S_0(\cdot)$ denotes the baseline survival function. The linear predictor η_i serves as an acceleration factor, meaning that covariates act to accelerate or decelerate time along the baseline survival function. However, estimating $S_0(\cdot)$ can be difficult, and we need to assume parametric models in practice. Therefore, we compared six candidate baseline distributions: Exponential, Weibull, Gamma, Log-logistic, Log-normal, and Generalized Gamma, using AIC and BIC as criteria. The Log-logistic distribution achieved the lowest AIC (653.0398) and BIC (690.4195). We therefore selected the Log-logistic distribution as our baseline for the final model.

Under the Log-logistic baseline distribution, the survival function takes the closed form shown below:

$$S(t | X_i) = S_0(te^{-\eta_i}) = \frac{1}{1 + \left(\frac{te^{-\eta_i}}{b}\right)^a}$$

where $a > 0$ is the shape parameter and $b > 0$ is the scale parameter, which allows for a non-monotone hazard function.

Note that the initial AFT model included all 9 covariates as main effects and with the same baseline distribution as the one used for the final AFT model (log-logistic).

We then applied several model diagnosis tools. We used the Cox-Snell residual plot to assess overall model fit. The Martingale residuals versus the numeric covariates were provided to check whether there's any sign of transformation of the numeric covariates. The plots of Martingale residuals and deviance residuals versus the linear predictor were used to identify potential model misspecification, specifically, whether the linear predictor is appropriate. Lastly, dfbetas versus each covariate were presented to identify any potential influential points.

For each fire in the dataset, we predicted the probability of a fire hitting an evacuation zone within 12, 24, 48, and 72 hours of initial detection using the fitted AFT model.

Model Evaluation. We evaluated model discrimination and calibration across all four prediction horizons using multiple approaches. The first is the time-dependent AUC, which measures the model's ability to discriminate between fires that have hit and those that have not at a given time point. The second metric is the IPCW Brier Score, which corrects for censoring bias through inverse probability of censoring weighting, and measures the calibration of predicted probabilities where lower values indicate better-calibrated predictions. The third metric is in-sample C-index and mean 3-fold cross-validated C-index.

Sensitivity Analysis. In this study, Random Survival Forest (RSF) was used as a sensitivity-analysis benchmark to assess whether the parsimonious AFT model might have overlooked meaningful nonlinear or interaction structure. RSF is a nonparametric ensemble tree method for right-censored survival data (9). It uses log-rank-type splitting and aggregates terminal-node Nelson–Aalen cumulative hazard estimators across trees to obtain an ensemble survival prediction. This mechanism allows the model to flexibly capture non-linearity and higher-order feature interactions without parametric survival-time assumptions and with reduced sensitivity to multicollinearity among correlated predictors.

To construct a comprehensive analysis, the RSF was initially trained on an expanded 17-variable exploratory set (see in previous Definition-based feature screening section). This broader set was used to assess whether variables removed from the primary AFT pipeline for interpretability or collinearity reasons masked stronger alternative predictive signals. For comparison, we also fit an RSF model using the reduced 8-variable feature set from the final AFT model. For both RSF models, raw continuous predictors were retained without additional transformation or standardization, because tree-based models are generally insensitive to monotone transformations and feature scaling.

Hyperparameters were tuned using a compact grid search and optimized using stratified 3-fold cross-validation based on the C-index criterion, including the number of trees, minimum leaf size, minimum split size, maximum tree depth, and the number of candidate features considered at each split. Given the limited sample size and number of observed events, we favored conservative tuning settings with relatively large minimum leaf sizes and restricted tree depth. Model performance was evaluated by in-sample C-index, mean 3-fold cross-validated C-index, and the gap between them as a descriptive indicator of apparent overfitting.

Results

This section presents the descriptive summary in the analytic sample, the model diagnostics that motivated transition from Cox to AFT, the final AFT results, and the sensitivity analyses used to assess robustness.

Descriptive Statistics. The analytic sample included 221 wildfire events, of which 69 reached within 5 km of an evacuation zone during follow-up and 152 were right-censored. The overall Kaplan–Meier curve showed substantial censoring and an early decline in survival probability, supporting the use of a time-to-event framework (see Figure 2).

Table 1. Descriptive statistics of raw covariates underlying the final AFT model, stratified by hit status during follow-up.

Feature	Fire Did Not Hit Within 72h (N=152)	Fire Hit (N=69)	Overall (N=221)
Low temporal resolution			
No	24 (15.8%)	36 (52.2%)	60 (27.1%)
Yes	128 (84.2%)	33 (47.8%)	161 (72.9%)
Initial area (log)			
Mean (SD)	4.92 (2.18)	4.17 (1.75)	4.68 (2.08)
Median [Min, Max]	5.10 [0.04, 9.39]	4.08 [0.30, 7.58]	4.71 [0.04, 9.39]
Centroid Movement Speed			
Mean (SD)	3.66 (25.87)	30.22 (96.23)	11.95 (58.94)
Median [Min, Max]	0.00 [0.00, 258.01]	0.00 [0.00, 595.06]	0.00 [0.00, 595.06]
Angle between fire move and evac (cos)			
Mean (SD)	-0.03 (0.28)	0.06 (0.52)	-0.00 (0.37)
Median [Min, Max]	0.00 [-1.00, 0.97]	0.00 [-0.96, 0.99]	0.00 [-1.00, 0.99]
Fire start hour			
Mean (SD)	15.18 (8.27)	15.99 (7.13)	15.43 (7.92)
Median [Min, Max]	19.50 [0.00, 23.00]	19.00 [0.00, 23.00]	19.00 [0.00, 23.00]
Fire start day of week			
Mean (SD)	3.00 (1.96)	2.49 (1.97)	2.84 (1.97)
Median [Min, Max]	3.00 [0.00, 6.00]	2.00 [0.00, 6.00]	3.00 [0.00, 6.00]
Fire start month			
Non-summer (1–5)	28 (18.4%)	8 (11.6%)	36 (16.3%)
Summer (6–9)	124 (81.6%)	61 (88.4%)	185 (83.7%)

Fig. 2. Kaplan–Meier survival curve.

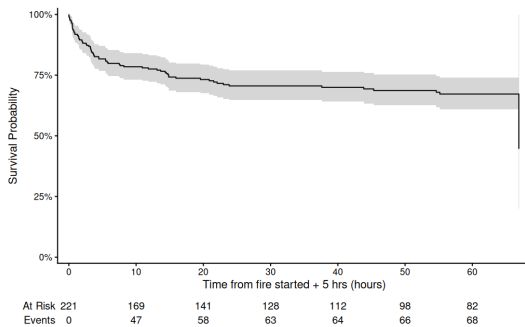


Table 2. Schoenfeld Residuals Test for Cox Proportional Hazards

Predictor	χ^2	df	p-value
Initial area (log)	4.093	1	0.043
Rate of radial growth (log)	1.811	1	0.178
Centroid Movement Speed (log)	1.788	1	0.181
Alignment Angle (cos)	1.016	1	0.313
Low temporal resolution	11.110	1	<0.001
Fire start hour (cos)	9.327	1	0.002
Fire start day of week (sin)	0.637	1	0.425
Fire start day of week (cos)	6.387	1	0.012
Summer indicator	0.638	1	0.424
GLOBAL MODEL	39.823	9	<0.001

Summary statistics for the raw covariates contained in the final AFT model are reported in Table 1, with additional exploratory plots provided in the Appendix. These summaries are shown on the original data scale to aid descriptive interpretation before preprocessing and feature engineering. Exploratory plots also suggested right-skewed wildfire-dynamics variables, cyclical temporal structure, and substantial correlation within several feature families, motivating the preprocessing and feature-reduction steps described in Materials and Methods Section.

Proportional Hazards Diagnostics. Schoenfeld residual tests for the initial 9-variable multivariable Cox PH model selected by Elastic Net indicated meaningful violations of the proportional hazards assumption (Table 2). Four predictors—initial area, low temporal resolution, hour cosine, and day-of-week cosine—showed significant evidence of violation, and the global test was also significant ($p < 0.001$). These results motivated transition from the Cox framework to the AFT model for primary inference.

Collinearity Refinement. The initial 9-variable AFT model showed severe collinearity between radial growth and centroid movement speed, with variance inflation factors (VIF) exceeding 100. We therefore compared two reduced models retaining only one of the two motion-related covariates (Table 3). Both reduced models showed nearly identical discrimination relative to the full model, while also improving AIC and BIC. We retained centroid movement speed in the final primary model. This decision was supported by slightly better AIC and BIC values together with a substantial reduction in collinearity (all VIFs < 1.7). More importantly, centroid movement speed is easier to interpret as a movement-related predictor for time-to-hit toward evacuation zones.

Table 3. Comparison of full and reduced AFT models under collinearity refinement.

Model	AIC	BIC	C-index	CV C-index
Full 9-variable AFT	653.040	690.420	0.773	0.758
Centroid speed-only	651.423	685.404	0.774	0.755
Radial growth-only	651.723	685.704	0.774	0.755

Final AFT Model Diagnosis. To assess the overall fit of the final AFT model, we examined Cox-Snell residuals, martingale residuals,

deviance residuals, and dfbetas. The Cox-Snell residual plot showed that the estimated cumulative hazard closely follows the 45-degree reference line for small to moderate residual values, suggesting an overall reasonable model fit. There is some deviation in the right tail (large residual values), but this is commonly observed in datasets with high censoring rates and is considered acceptable in our case. Martingale residual plots against numeric predictors were mostly flat around zero with no obvious patterns, indicating that the linear form for most predictors is appropriate. There were some curvature for log centroid speed and alignment cosine, but is reasonable considering their zero-inflation characteristics. Martingale and deviance residuals plotted against the linear predictor showed mild curvature, suggesting potential but limited model misspecification. Finally, dfbeta plots did not reveal any highly influential observations.

AFT Model Estimates. The estimated coefficients from our fitted final AFT model can be seen in Table 4, and the 95% CI were also provided.

Table 4. Final 8-variable AFT model estimates under a log-logistic distribution.

Covariate	Estimate	SE	95% CI
Log Initial Area	0.240	0.292	(-0.332, 0.812)
Log Centroid Speed	-0.507*	0.245	(-0.988, -0.026)
Alignment Angle (cos)	-0.323	0.212	(-0.738, 0.093)
Hour (cos)	0.289	0.249	(-0.199, 0.777)
Low Temporal Resolution (Yes)	3.277*	0.684	(1.936, 4.617)
Day of Week (sin)	-0.565*	0.275	(-1.105, -0.025)
Day of Week (cos)	-0.282	0.267	(-0.805, 0.242)
Summer (Yes)	-0.671	0.784	(-2.208, 0.865)

* $p < 0.05$. Overall model significance test: $\chi^2_8 = 66.86$, $p < 0.001$.

As indicated by the negative estimated coefficient, fires with higher centroid speed were significantly associated with a shorter time to hit the evacuation zone, meaning fires with faster centroid movement reach evacuation zones more quickly (estimate = -0.507, 95% CI: [-0.988, -0.026]). Similarly, the sine of day of week showed a significant negative association, suggesting temporal variation in fire risk across the week (estimate = -0.565, 95% CI: [-1.105, -0.025]). In contrast, low temporal resolution in early fire observations was significantly associated with a longer time to hit (estimate = 3.277, 95% CI: [1.936, 4.617]).

Predicted Probability of Fire Hit. Table 5 displayed the predicted hit probabilities for four representative fires across four time horizons: 12, 24, 48, and 72 hours after fire ignition. Note that for the 72-hour probability prediction, since the maximum observed follow-up time was slightly below 67 hours after the landmark time (66.99 hours), the prediction time was set to this maximum follow-up time.

Since we predicted for all 221 observations/fires in the dataset, we only presented the results for the first four of them, considering the length of the predicted table. Taking Fire 3 as an example, it did hit with a time to hit of only 0.89 hours after the landmark time. Even at the 12-hour horizon, the model assigns a hit probability of 0.331, rising to 0.653 at 72 hours, showing that the model can identify high-risk fires relatively well. In contrast, Fire 4 was censored with a follow-up time of 60.95 hours, and the model assigns consistently low probabilities across all horizons, peaking at only 0.155 at 72 hours, indicating reasonable classification for low-risk cases. Other than this, an interesting comparison is between Fire 1 and Fire

2. Both fires have similar follow-up times (18.89 and 22.05 hours, respectively), and the model assigns similar predicted probabilities, for example, 0.373 versus 0.360 at the 24-hour horizon, while Fire 1 did not hit the evacuation zone and Fire 2 did hit. This reveals a potential limitation of our model on identifying the risk between fires that may share similar early-stage characteristics.

It is also important to note that all predicted probabilities here are conditional on surviving past the landmark time. That is, they represent the probability of hitting given that the fire was still active at 5 hours after initial perimeter detection, not the absolute risk from ignition. This conditional formulation should therefore be interpreted as landmark-based risk rather than absolute ignition-time risk.

Table 5. Predicted cumulative probability of evacuation-zone hit at four operational horizons

ID	Fire Hit	Time to Hit (h)	Horizon	Est. Risk	95% CI
1	No	18.893 + 5	12h	0.248	(0.096, 0.507)
1	No	18.893 + 5	24h	0.373	(0.167, 0.655)
1	No	18.893 + 5	48h	0.491	(0.243, 0.751)
1	No	18.893 + 5	72h	0.556	(0.295, 0.807)
2	Yes	22.048 + 5	12h	0.238	(0.120, 0.432)
2	Yes	22.048 + 5	24h	0.360	(0.196, 0.590)
2	Yes	22.048 + 5	48h	0.477	(0.293, 0.677)
2	Yes	22.048 + 5	72h	0.542	(0.339, 0.751)
3	Yes	0.889 + 5	12h	0.331	(0.162, 0.583)
3	Yes	0.889 + 5	24h	0.472	(0.254, 0.692)
3	Yes	0.889 + 5	48h	0.592	(0.366, 0.804)
3	Yes	0.889 + 5	72h	0.653	(0.414, 0.843)
4	No	60.953 + 5	12h	0.046	(0.022, 0.090)
4	No	60.953 + 5	24h	0.080	(0.040, 0.146)
4	No	60.953 + 5	48h	0.123	(0.066, 0.223)
4	No	60.953 + 5	72h	0.155	(0.086, 0.279)

Model Evaluation. Table 6 showed the results of model evaluation via AUC and Brier Score. In terms of AUC, the model exceeds 0.80 across all four horizons, reaching 0.861 at 12 hours and 0.876 at 72 hours, indicating strong discriminative performance throughout the follow-up window. For the Brier Score, the AFT model consistently performs better than the reference baseline model (KM estimate), confirming that incorporating early-stage fire characteristics improve predictive performance. It's worth noting that since 72 hours is the maximum follow-up time in our dataset, the risk set at that point is smaller, and the censoring proportion is higher, meaning that the 72-hour evaluation may be less stable than earlier horizons. In general, based on these two methods, our model performs reasonably well to rank fires by their relative risk, which means fires that eventually hit tend to be assigned higher predicted probabilities than those that do not, at each given time horizon.

Table 6. Model evaluation metrics across prediction horizons

Horizon	AUC	Brier Score (Reference)	Brier Score
12h	0.861	0.161	0.106
24h	0.804	0.194	0.144
48h	0.807	0.210	0.163
72h	0.876	0.247	0.223

Predicted probabilities are conditional on the 5-hour landmark. AUC = Area Under the ROC Curve. Brier Score (IPCW) measures calibration (lower is better). Reference denotes the baseline model (KM estimate).

Sensitivity Analysis. We conducted sensitivity analyses to assess whether a more flexible nonlinear model or alternative directional encodings would materially change the main conclusions from the primary AFT model.

Random Survival Forest Results. As a non-linear sensitivity benchmark, Random Survival Forest achieved a cross-validated discrimination that was only marginally better than the primary AFT model.

In the expanded 17-variable RSF, the permutation importance highlighted directional alignment, fire-start day-of-week, initial fire area, and observation quality as the strongest signal domains. This indicates that the main signals were already captured by the AFT model. As shown in Table 7, RSF reached a mean 3-fold CV C-index of 0.7671 using the expanded feature set, whereas it reached a nearly identical value of 0.7667 using the reduced AFT-based feature set, suggesting that dimensionality reduction preserved most of the main predictive signals.

Furthermore, while RSF improved in-sample fit, it provided only modest improvement in cross-validated performance with much larger train–CV performance gaps (0.059–0.060). This may suggest probable overfitting despite conservative tuning. In contrast, AFT achieves a much tighter train–CV gap (0.019) with a competitive cross-validated performance (C-index = 0.755). Overall, RSF showed only modest nonlinear predictive gains, supporting the continued use of the simpler and more interpretable AFT model as the primary inferential framework.

Table 7. Internal performance comparison between the primary AFT model and RSF model

Model	In-sample C-index	CV C-index	Gap
AFT log-logistic	0.774	0.755	0.019
RSF (AFT Features)	0.826	0.767	0.060
RSF (Expanded Features)	0.825	0.767	0.057

Alternative Directional Variables. Since expanded RSF identified the absolute cosine value of alignment angle (`alignment_abs`) as the top predictor, we compared alternative directional encodings in the AFT model. Replacing `alignment_cos` with `alignment_abs`, or including both together, failed to yield meaningful improvements (Table 8). Our final model (cosine-only specification) maintained a balance in AIC, BIC, and cross-validated stability. Therefore, the alignment cosine is retained in the final primary model, as it preserves directional sign (toward versus away from evacuation zones), whereas the absolute version collapses this distinction into unsigned alignment magnitude.

Table 8. Alternative directional-variable comparison in AFT models

Model	AIC	BIC	C-index	CV C-index
only cos (Final)	651.423	685.404	0.774	0.755
Both (cos + abs)	653.324	690.703	0.775	0.742
only abs	653.550	687.531	0.769	0.741

Exploratory Nonlinear Checks. Exploratory partial dependence plots of RSF (see Appendix) and spline-based AFT checks suggested only limited additional nonlinear signals. These findings suggest that the complex non-linear terms would not yield sufficient predictive gain.

Discussion and Conclusion

Key findings. Our results suggest that short-term wildfire risk is more related to fire dynamics, fire-start time, and the observation process than to static geometric characteristics.

First, fire centroid movement speed appears more relevant to time-to-hit than static initial area. After refining highly correlated fire growth features, centroid movement speed remained the more stable (smallest SE) and interpretable dynamic signal in the final AFT model, suggesting that faster-moving fires tend to reach evacuation zones sooner. By contrast, initial fire area was not a significant predictor once movement speed and directional alignment are controlled, providing limited additional information about when it will reach a specific location. Directional alignment did not show significance in the final AFT model, but it remained as one of the strongest predictors in the RSF sensitivity analysis. This suggests that directionality toward evacuation zones could remain part of the broader predictive signal.

Second, temporal context remained informative. A cyclical day-of-week effect was significant in the final AFT model, suggesting that short-term risk may vary with certain time period, which may reflect anthropogenic activity. However, given the limited sample size, this finding should be interpreted cautiously. By contrast, the summer indicator was not statistically significant in the final model. This may indicate that some seasonal variation was already captured by early fire kinematic variables. This is consistent with prior work: Tremblay et al. also retained weather and fuel-type variables rather than a seasonal indicator (6), suggesting that immediate short-term conditions (e.g., fire kinematics or weather) may be more informative than long-term seasonal effects.

Third, the strong effect of the low-resolution flag suggests that the observation process itself is informative in this dataset. This variable may reflect differences in operational attention or underlying fire severity rather than observation quality. Therefore, the low-resolution flag should not only be treated as a control variable, but also interpreted as evidence that wildfire risk may be influenced by informative observation processes.

Our findings align with prior wildfire survival work, which has emphasized fire characteristics at initial assessment and external factors such as weather, fuel type, and management conditions (5, 6). In our dataset, the kinematic variables may partly reflect these factors. However, our study design differs because the outcome is not fire suppression, but time to evacuation-zone exposure within a short follow-up window. In this urgency framework, fire dynamics become more essential, whereas static initial area and seasonal effects may contribute less. The contrast with prior work, where initial fire size was important, likely due to the difference in outcome variable: control time and fire-size growth measure overall suppression difficulty and fire geometry, whereas time-to-hit is fundamentally a directional and kinematic outcome. This helps explain why initial fire area was less significant here than in prior wildfire survival studies. The results of this time-to-hit framework provide recommendations for evacuation planning and emergency response implementation.

From a practical point of view, our results suggest that a landmark-based survival framework and early-stage characteristics can provide useful short-time urgency discrimination. Rather than treating the problem as a binary fire hit outcome, the time-to-hit framework offers a meaningful ranking of how quickly and how likely a fire may reach nearby evacuation zones. Compared to physical fire-spread models that require extensive computational

resources and detailed spatial inputs (2, 3), our approach relies on easily accessible early-stage characteristics. While it is not a replacement for physical simulation models, this type of conditional risk prediction may support prioritization and resource allocation, especially when decision-makers face multiple simultaneously threatened zones under time pressure. In full-sample evaluation, the parsimonious AFT model achieved AUC values above 0.80 across all time windows, suggesting that early-stage urgency ranking may be feasible. Its mean 3-fold cross-validated concordance index was 0.7552, indicating moderate but potentially useful discrimination under internal validation. This has practical significance for evacuation planning and emergency situation classification, as it can provide actionable early warning information before more complex physical simulations are completed.

Limitations. Several limitations should be noted. First, the low-resolution flag likely captures more than observation quality alone and may partly act as a proxy for fire severity, indicating potential observation bias. In particular, low-resolution fires may reflect differences in detectability or operational attention. Second, the 5-hour landmark design conditions the analysis on fires still event-free at the landmark and therefore estimates conditional risk from that landmark time rather than absolute risk from ignition. Third, the outcome may also be influenced by administrative factors: evacuation-zone boundaries depend not only on fire behavior but also on local policies and operational practices, which were not accounted for in this analysis. Moreover, the dataset includes intrinsic fire-characteristic variables, but limited exogenous context. Important drivers of fire spread, such as location, weather, terrain, and vegetation, are absent, which likely constrains predictive performance. Finally, the current primary model includes linear terms only. Although AFT spline extensions and RSF yielded little additional predictive gain, this should not be interpreted as proof that nonlinear structure is absent; it may instead reflect limited sample size and incomplete contextual information.

Future work could include adding more relevant information on fire spread, as well as exploring regional differences and spatial variation in how evacuation zones are defined, where data are available. It would also be helpful to test the model on external datasets to better assess how well it generalizes. In addition, instead of relying on a single fixed landmark, future work could extend this approach to a dynamic landmarking framework that updates predictions over time (10). This would allow continuous, real-time updates of evacuation risk as fire behavior evolves during an active event. Ultimately, this could support a more flexible early-warning tool for ongoing emergencies.

Appendix

This appendix provides supporting tables and exploratory figures referenced in the main text, including variable definitions, expanded RSF feature families, and partial dependence plots.

Variable Dictionary for Final AFT Model. See Table 9.

Expanded RSF Feature Families. See Table 10.

Exploratory Data Analysis.

Exploratory partial dependence plots.

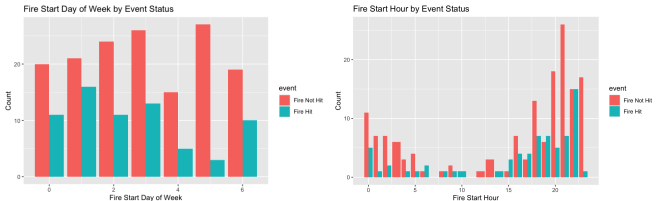


Fig. 3. Left: histogram of fire-start day of week. Right: histogram of fire-start hour.

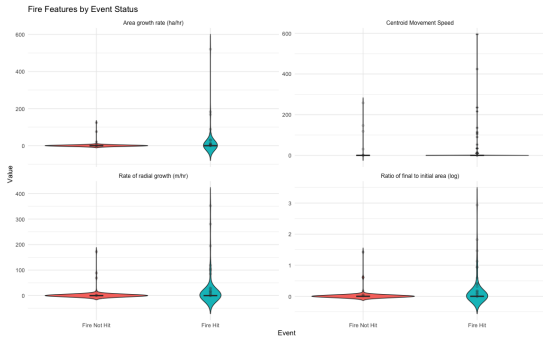


Fig. 4. Violin and box plots for selected raw wildfire-dynamics variables before transformation, illustrating strong right-skewness that motivated log transformation.

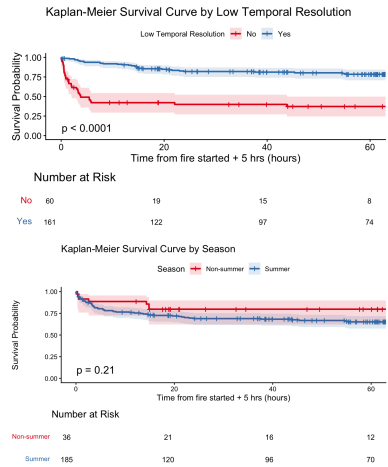


Fig. 5. Top: Kaplan–Meier curve for the low-resolution subgroup. Bottom: Kaplan–Meier curve for the summer subgroup.

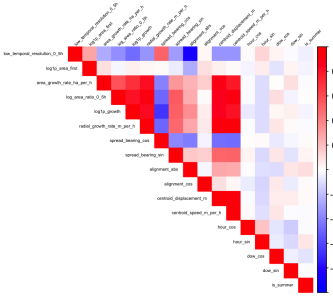


Fig. 6. Correlation heatmap for the 17-variable exploratory variable set used before AFT-specific refinement.

Table 9. Concepts, variable names, and brief definitions for the final 8-variable AFT model.

Concept	Variable name	Brief definition
Observation quality	low_temporal_resolution_0.5h	Equals 1 if only one observation is available in 0–5 hours or if the observation time span is less than 0.5 hours.
Initial fire area	log1p_area_first	Log-transformed initial fire area (hectares).
Centroid movement speed	log_centroid_speed	Log-transformed centroid movement speed of the fire perimeter (meters/hour).
Directional alignment angle	alignment_cos	Cosine of the alignment angle between fire movement direction and evacuation-zone direction.
Fire start hour	hour_cos	Cosine-encoded fire-start hour.
Fire start day of week	dow_sin, dow_cos	Cyclic encoded fire-start day-of-week.
Fire start season	is_summer	Binary summer indicator; equals 1 when fire starts between June and September.

Table 10. Expanded RSF feature families used in the 17-variable sensitivity-analysis benchmark.

Feature family	Variables
Observation quality	low_temporal_resolution_0.5h
Fire geometry	log1p_area_first, log1p_growth, log_area_ratio_0.5h, area_growth_rate_ha_per_h, radial_growth_rate_m_per_h
Fire kinematic	centroid_speed_m_per_h, centroid_displacement_m
Direction	alignment_cos, alignment_abs, spread_bearing_sin, spread_bearing_cos
Fire start time	hour_sin, hour_cos, week_sin, week_cos, is_summer

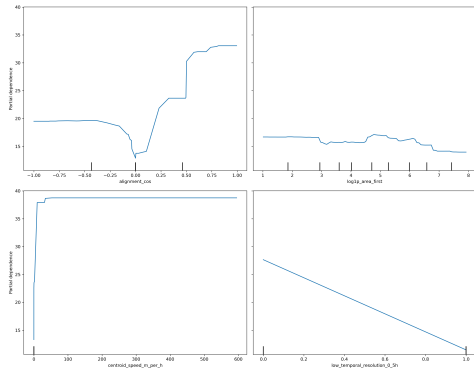


Fig. 7. Exploratory partial dependence plots from the RSF sensitivity analysis.

Data, Materials, and Software Availability. All datasets and codes required to reproduce this study are included in the submission. The primary Accelerated Failure Time (AFT) modeling and exploratory data analyses were implemented in R v.4.5.3, provided in the R Markdown document `Final_result_revised.Rmd`. The non-parametric Random Survival Forest (RSF) sensitivity analysis was executed in Python and is included as the Jupyter Notebook `Sensitivity_Random_Survival_Forest.ipynb`.

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